🔑 **Answer Key — Teacher Copy · Lesson 8-7**

MSTAR PRACTICE WORKSHEET

Lesson 8-7: Equations and Inequalities Problem Solving

Grade 6 Mathematics · Standard 6.AT.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

A museum has some paintings. After adding 12, they have 45. Which equation models this?

- **A** $p + 12 = 45$
- **B** $p - 12 = 45$
- **C** $12p = 45$
- **D** $p / 12 = 45$

'Adding 12' to the original number gives 45: $p + 12 = 45$. Solve: $p = 33$.

Question 2 SELECTED RESPONSE — CORRECT: A

A detective needs more than 8 hours to finish the investigation. She has already worked 3 hours. Which inequality represents the additional hours h she needs?

- **A** $3 + h > 8$
- **B** $3 + h = 8$
- **C** $h > 3$
- **D** $8 + h > 3$

3 hours plus additional hours h must be more than 8: $3 + h > 8$. Solve: $h > 5$.

Question 3 SELECTED RESPONSE — CORRECT: A

Which model is correct for: 'A number divided by 6 equals 7'?

- A $n / 6 = 7$
- B $6n = 7$
- C $n - 6 = 7$
- D $n + 6 = 7$

> Divided by 6 is n / 6. Equals 7 means = 7. So n / 6 = 7.

Question 4 SELECTED RESPONSE — CORRECT: A

A bus can carry at most 48 passengers. There are already 31 on board. Which inequality shows how many more can board?

- A $31 + p \leq 48$
- B $31 + p > 48$
- C $31 + p = 48$
- D $p - 31 \leq 48$

> 'At most 48' means the total must be ≤ 48. So $31 + p \leq 48$. Solve: $p \leq 17$.

Question 5 SELECTED RESPONSE — CORRECT: C

A student solved $4n = 52$ and got $n = 13$. Is the answer reasonable if n represents the number of notebooks in a box?

- A No — 13 is too many notebooks
- B No — the answer should be 48
- C Yes — 13 notebooks per box is reasonable
- D Yes — but only if n is a fraction

> $4 \times 13 = 52$ ✓. 13 notebooks per box is a reasonable whole number answer.

Question 6 SELECTED RESPONSE — CORRECT: C

Room 2: The vault keypad clue reads, 'A number of stolen coins plus 6 equals 14.' Solve $x + 6 = 14$ to find the code x.

- (A) $x = 2$
- (B) $x = 6$
- (C) $x = 8$
- (D) $x = 20$

> Subtract 6 from both sides: $x = 14 - 6 = 8$. Check: $8 + 6 = 14$ ✓

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A school van can carry at most 14 passengers. There are already 9 passengers on board. Let p be the number of additional passengers who can get on. Which inequality models this situation?

- A $9 + p \le 14$
- B $9 + p \ge 14$
- C $9 + p < 14$
- D $p - 9 \le 14$

> The 9 passengers already on board plus the p additional passengers make up the total. The phrase 'at most 14' means that total may reach 14 but cannot pass it, so the model is $9 + p \le 14$.

- **B:** This demands at least 14, which would let the van overfill. 'At most' caps the total.
- **C:** This forbids a full van of exactly 14. 'At most 14' allows 14.
- **D:** This subtracts the riders already aboard. They add to the total, so 9 plus p.

Part B — correct answer: C

Which reasoning justifies that model and gives the greatest number of additional passengers?

- A The van must end up completely full, so $9 + p = 14$ and exactly 5 more passengers must get on.
- B The 9 passengers are taken away from p, so $p - 9 \le 14$ and at most 23 more passengers can get on.
- C The total cannot pass the limit of 14, so $9 + p \le 14$ and at most 5 more passengers can get on.
- D The words 'at most' mean the total has to be more than 14, so $9 + p \ge 14$ and 5 or more passengers can get on.

> 'At most 14' sets a maximum the total may reach but not pass, so $9 + p \le 14$. Subtracting 9 from both sides gives $p \le 5$. Checking in context confirms it: $9 + 5 = 14$ fits the van, while $9 + 6 = 15$ is over the limit. The situation does not require the van to be full, so an equation would leave out the smaller totals that are also allowed.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C

A club has already sold 46 tickets and must sell at least 120 tickets in all. Let t be the number of additional tickets the club sells. Select ALL statements that are true about this situation.

- A The inequality $46 + t \geq 120$ models the situation.
- B The club must sell at least 74 more tickets to reach its goal.
- C Selling exactly 74 more tickets reaches the goal.
- D The inequality $46 + t \leq 120$ models the situation.
- E Selling 70 more tickets reaches the goal.

The tickets already sold plus the additional tickets must reach at least 120, so the model is $46 + t \geq 120$. Subtracting 46 from both sides gives $t \geq 74$. Selling exactly 74 more works because $46 + 74 = 120$. The fourth statement points the symbol the wrong way, since 'at least' sets a minimum, not a maximum. The fifth is false because $46 + 70 = 116$, which is short of 120.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Create two word problems about the same situation: one that requires an equation and one that requires an inequality. Solve both and explain why the models are different.

Model answer: The two models differ because the two questions ask for different things. An equation fits when the situation names one exact amount, like 'I saved exactly $45', so it has a single solution. An inequality fits when the situation sets a minimum or a ceiling, like 'I need at least $45', so a whole range of values works. Same story, different clue words, different symbol.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.